

Insurance Policy Analysis

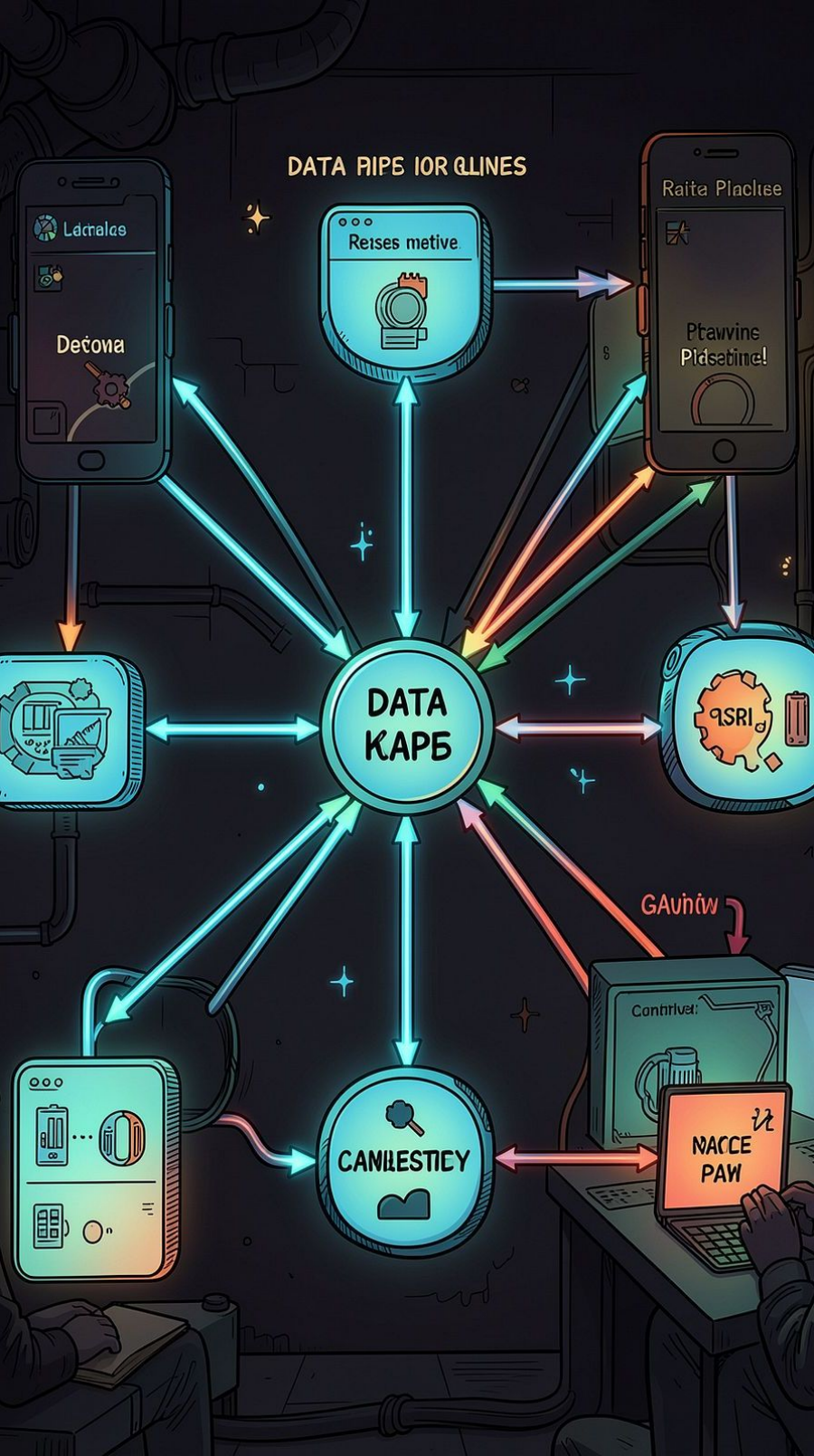
End-to-End Data Analytics Project — transforming raw insurance data into meaningful business insights.

PYTHON

SQL SERVER

POWER BI





Project Overview

This project analyzes insurance policies, customer information, premiums, coverage amounts, and claims through a complete analytics workflow – from raw data to interactive dashboard.

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Data Loading

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Cleaning & EDA

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Power BI Dashboard

Project Objectives

Data Understanding

Explore dataset structure, characteristics, and data quality.

Policy & Claim Analysis

Analyze premiums, coverage, claim patterns, and policy status.

Customer Segmentation

Study customer age groups, gender, and policy type distributions.

Business Intelligence

Perform SQL queries and build an interactive Power BI dashboard.



Dataset at a Glance

The dataset contains ~5,001 insurance policies with fields including Policy Number, Customer ID, Policy Type, Premium, Coverage, Claim Amount, Claim Status, Dates, Age Group, and Gender.

5,001

Total Policies

2.99M

Total Premium

301.63M

Total Coverage

8.41M

Total Claims

2.9K

Active Policies

2.1K

Inactive Policies

Tools & Technologies

Python

Pandas, NumPy, Matplotlib, Seaborn – data loading, cleaning, EDA, and visualization.

SQL Server

Filtering, aggregation, grouping, and structured business queries.

Power BI

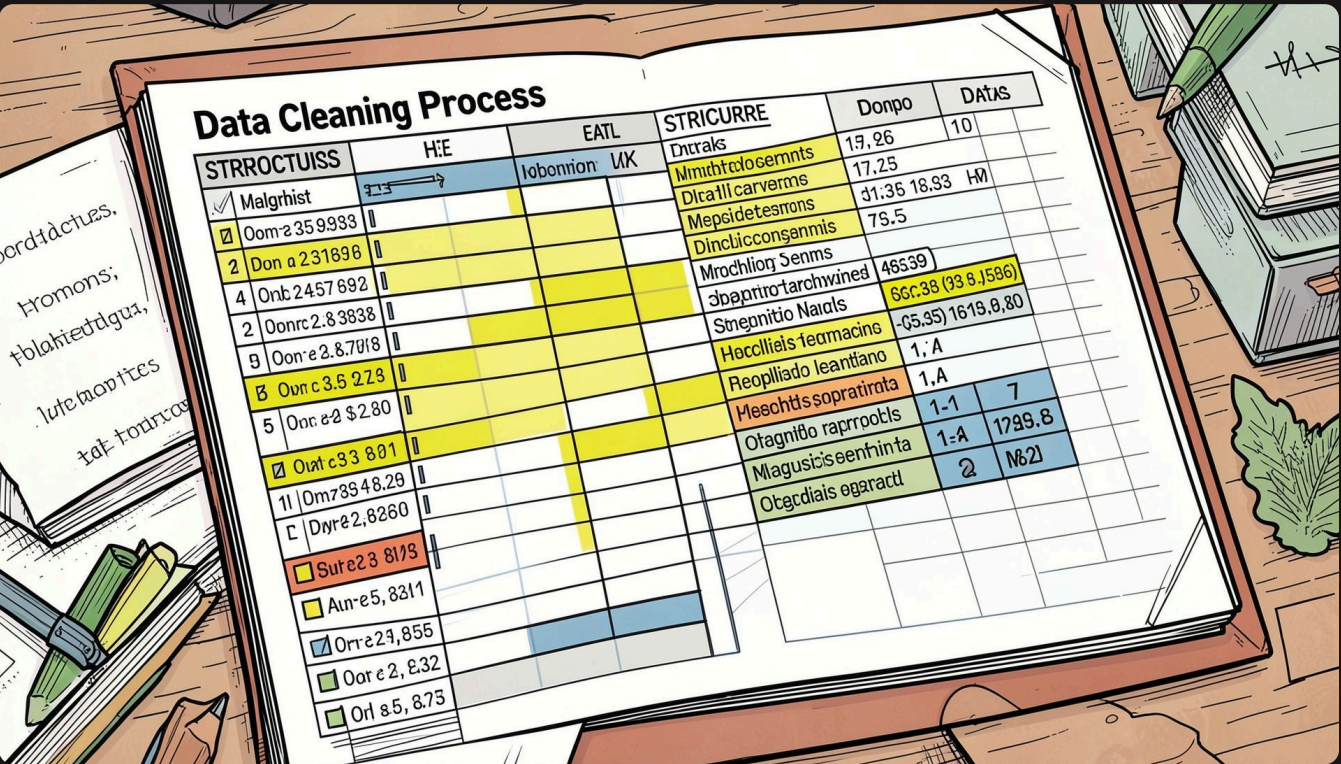
Interactive dashboard with KPIs, charts, filters, and category-level analysis.

CSV / Excel

Structured source format for the raw insurance dataset.



Data Loading & Cleaning



Preprocessing Steps

01

Inspect Structure

Check rows, columns, data types, and missing values using `isnull()` and `sum()`.

02

Handle Missing Values

Apply median replacement or removal depending on the variable.

03

Remove Duplicates

Eliminate redundant records to ensure data integrity.

04

Standardize Formats

Convert date columns and validate numerical and categorical fields.

Exploratory Data Analysis

What We Explored

- **Numerical Variables:** Premium, Coverage, Claim Amount, and Customer Age analyzed via descriptive statistics and distribution plots.
- **Customer Demographics:** Age groups and gender used to segment and compare the customer base.
- **Policy Types:** Distribution across products compared to understand portfolio composition.
- **Claim Patterns:** Claim counts, amounts, and status analyzed across policy categories.
- **Policy Status:** ~2.9K active vs. ~2.1K inactive policies examined.



Statistical Analysis & SQL Queries

Correlation Analysis

A correlation matrix and heatmaps investigated relationships between **Customer Age**, **Premium**, **Coverage**, and **Claim Amount**. Positive correlations indicate variables that increase together; negative correlations show inverse relationships. Correlation represents association, not causation.

SQL Business Questions



Total policies, premiums, and coverage amounts



Claims distribution by claim status



Active vs. inactive policy counts



Premiums and claims across policy types and age groups

Power BI Dashboard & Key Findings

5,001 Policies

Total portfolio size

2.99M Premium

Total premiums collected

301.63M Coverage

Total coverage amount

8.41M Claims

Total claims paid

The interactive dashboard enables visual analysis across policy type, status, claim status, age group, and gender – with filters to explore specific segments. Key findings: ~2.9K active vs. ~2.1K inactive policies, and policy-type comparisons reveal distinct premium, coverage, and claim patterns across the portfolio.



Challenges, Conclusion & Future Scope

Challenges Faced

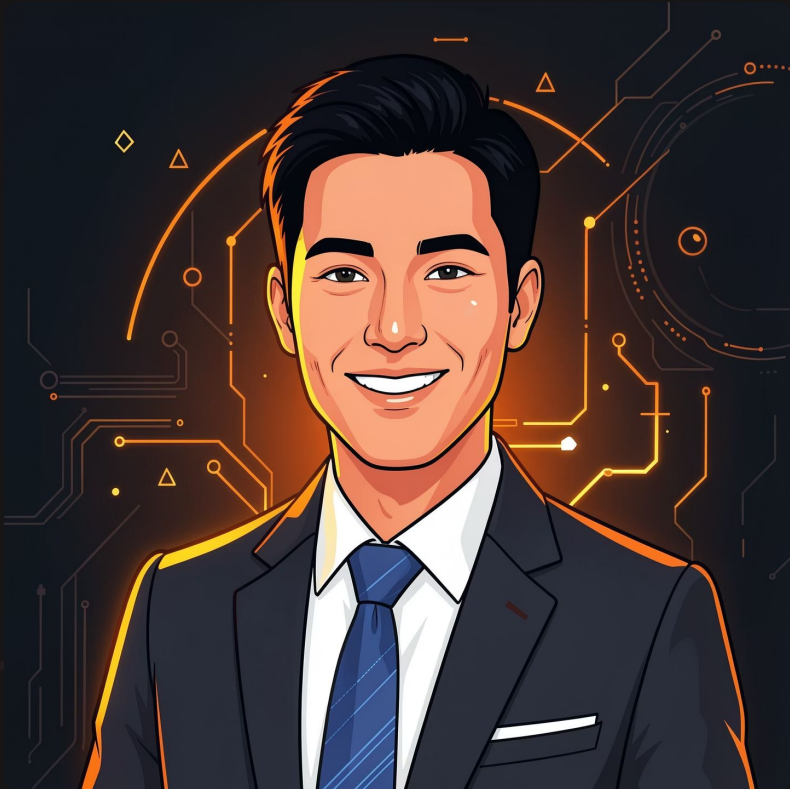
- **Data Quality:** Missing values, inconsistent formats, and incorrect data types required careful preprocessing.
- **Data Interpretation:** Premium, coverage, and claims represent different aspects of insurance performance.
- **Visualization:** Selecting clear, effective visuals without overcomplicating the dashboard.
- **Multi-Tool Integration:** Coordinating Python, SQL Server, and Power BI.

Future Scope

- Predictive models for claim risk and customer churn
- Time-series analysis of claims and premiums
- Automated Power BI refresh pipelines
- Advanced customer segmentation and ML-based claim prediction

📄 This project demonstrates end-to-end analytics – from raw data to business intelligence – using Python, SQL Server, and Power BI.

About the Author



Siddharth Pathak

Siddharth Pathak is a skilled Data Analyst with expertise in transforming complex datasets into actionable business insights. He specializes in end-to-end data analytics projects, leveraging powerful tools to drive data-driven decision-making and enhance business performance.

POWER BI

SQL

EXCEL

PYTHON

EDA

PANDAS

NUMPY